

CSE2315 Lab Course

Assignment 5

- This assignment consists of several exam-level questions and sometimes an optional extra exercise.
- See Brightspace, *Course Information* for instructions about the lab course, rules and grading procedure.
- Your solution has to be handed in on paper, typeset in $\text{\LaTeX}$, using a word processor or readable handwriting. Handing in via email is not permitted.
- Cooperating with a colleague is permitted and *strongly encouraged!* In such cases, please hand in a single copy of the work.
- Total number of pages, without this cover page: 1.

1. Consider the alphabet $\Sigma = \{a, b, c\}$ and the language

$$L = \{a^k b^k c^k \mid k \geq 1\}.$$

Give a high-level description for a three-tape enumerator that enumerates L . You may want to use the second and third tapes as counters. You do not have to give implementation details of the counters.

2. *Old exam question endterm 23-24:*

- (a) Give a Turing Machine M that is not a decider such that $L(M)$ is decidable.
- (b) Explain why M is not a decider.
- (c) Explain why $L(M)$ is decidable.

3. *Old exam question resit 20-21:* Let $MIA_{TM} = \{\langle M_1, M_2 \rangle \mid L(M_1) \cap L(M_2) \neq \emptyset\}$. Show that this is a recognizable language.

4. In the game of chess, consider an intermediate board configuration, and suppose it's white's move. We know that white is either guaranteed to win or not. Let

$$L = \{w \mid w \text{ represents a board configuration, and white is guaranteed to win if it is white's move and white plays optimally}\}.$$

Is L recognizable, decidable, or neither? Explain.

5. *Revision, Old exam question, midterm 21-22:* Given a DTM M with $\Sigma = \{a, b\}$ such that:

- M has no transition to its q_{reject} i.e. the reject state is an unreachable state
- $L(M) = \{w \in \Sigma^* \mid w \text{ starts with an } a\}$

Is M a decider, not a decider, or can we not conclude that? Explain your answer.

6. *Old exam question resit 20-21:*

In a log of a scheduler, it is noted down when a request for a task is given, and when an action is done to do that task. Requests are noted down with r , actions with a . A log is a final log if all tasks have been done. Thus, a final log can be described by the language of words where every r has a corresponding a that appears later in the string, formally given as $L = \{w \mid w \text{ has an equal amount of } r\text{'s and } a\text{'s and the } i^{\text{th}} a \text{ appears later in the string than the } i^{\text{th}} r, \text{ for each } i \text{ in } (1, |w|/2)\}$. Give an informal description of a Turing machine with the alphabet $\{r, a\}$ that decides this language. Use a maximum of 15 lines.

As an example: $rarraa \in L$ and $raaarr \notin L$

7. *Revision, old exam question midterm 19-20:* Consider the following rules R of a CFG $G = (V, \Sigma, R, S)$, with $V = \{S, A, B\}$:

$$S \rightarrow ASA \mid Ba$$

$$A \rightarrow bB \mid S$$

$$B \rightarrow c \mid b \mid \varepsilon$$

- (a) (1 point) Give a valid set Σ such that $|\Sigma| = 5$.
- (b) (6 points) Convert G into Chomsky Normal Form (CNF). Use the procedure from Sipser, showing intermediate steps.

Sorry, no optional extra exercise puzzle this week. But are you up to date with the Quadruple Quest?